From: Víctor Casaña-Giner, Universidad Nacional de Educación a Distancia, Madrid, Spain
vcasana1@alumno.uned.es ORCID 0000-0002-8252-2608

To: Victor Amberger, Chiara Bellotti, Peng-Jie Wong

Subject: Optimised kernels and a hybrid variant of your Turing-type method for $S(T)$

Date: 31 October 2026

Dear Dr. Amberger, Dr. Bellotti and Professor Wong,

I have read with great interest your preprints [1] and [3], and earlier [2]. I write to tell you what I have obtained by building on your Turing-type method [8], how I obtained it, and how it compares with your results. The method is yours; my contributions are an optimisation of the kernel, a sharper treatment of the constants, and a hybrid variant for the asymptotic coefficient. I would be grateful for any comments before I write these results up.

Throughout, $N(T)$ counts the zeros $\rho = \beta + i\gamma$ of ζ with $0 < \gamma \le T$, $\mathrm{M}(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e}$, $L = \log T$, and $\mathfrak{a} = \limsup_{T\to\infty} |S(T)|/\log T$. All numerical inequalities below are certified in Arb ball arithmetic [6] at 256 bits. No numerical data on the zeros (in particular no computation of Platt) is used. The only arithmetic input beyond the Euler product is $\psi(x) < 1.03883x$ [7, Thm. 12].

1. What I obtained

(a) Zero counting, all $T \ge 1$. For every $T \ge 1$,

$$|N(T) - \mathrm{M}(T)| \le \min\{0.20176L + 1.505,\ 0.10823L + 2.382,\ 0.09473L + 3.075,\ 0.09349L + 3.237,\\ 0.09152L + 3.776,\ 0.09012L + 4.593,\ 0.08844L + 6.988\}. \tag{1.1}$$

The seven lines are active on $L \in [0, 9.4]$, $[9.4, 51.3]$, $[51.3, 130.6]$, $[130.6, 273.6]$, $[273.6, 583.6]$, $[583.6, 1425.6]$, $[1425.6, \infty)$. The right side of (1.1) lies below $\min(0.0946L + 5.746,\ 0.097L + 4.954)$ of [3, Cor. 1.3] for every $T \ge 1$. The smallest difference is 1.996, at $L \approx 51.4$, and the difference tends to infinity. Each of the lines $0.09349L + 3.237$ and $0.09012L + 4.593$ is, on its own, below both of your lines for all $T \ge 1$.

(b) The argument, all $T \ge 1$. With the same seven kernels,

$$|S(T)| \le \min\{0.20176L + 0.979,\ 0.10823L + 1.498,\ 0.09473L + 2.187,\ 0.09349L + 2.350,\\ 0.09152L + 2.887,\ 0.09012L + 3.704,\ 0.08844L + 6.099\}. \tag{1.2}$$

For comparison, [2, Cor. 1.5] gives $0.10076L + 0.24460\log L + 7.20844$, and [3, Cor. 1.3] gives $|S(T)| \le 0.0946L + 6.691$ for $T \ge e$, through $|S| \le |N - \mathrm{M}| + \frac{7}{8} + |\varpi|/\pi$.

(c) Your own kernels. With the tuples (2.43) and (2.44) of [3] (rounded to binary) and the treatment of the constants of Section 2,

$$|N(T) - \mathrm{M}(T)| \le \min\{0.0946L + 5.138,\ 0.097L + 4.507\} \qquad (T \ge 1), \tag{1.3}$$

in place of 5.746 and 4.954.

(d) The asymptotic coefficient. A hybrid of your kernel majorant with Jensen discs, described in Section 2, gives

$$\mathfrak{a} < 0.0775422804104 < \Lambda(0) = \frac{3\sqrt{3} - \pi}{12\pi\log 2} = 0.0786252\ldots, \tag{1.4}$$

where $\Lambda(0)$ is the value that Backlund's method gives under the Lindelöf hypothesis. This bound is asymptotic only: no bound at any finite height is claimed. On a grid, without certification, the same construction gives about 0.0760 (kernel of order 10, 1045 discs) and 0.0752 (order 20).

2. How

The identity. As in [1, §2] and [3, §2], for $d > \frac{1}{2}$ and T not an ordinate,

$$N(T) = \frac{1}{\pi}\Big(2\arg\xi(\tfrac{1}{2}+d+iT) - \arg\xi(\tfrac{1}{2}+2d+iT)\Big) + \frac{1}{2\pi}\sum_{\rho} f(\tfrac{1}{2}-\beta,\, T-\gamma), \qquad (2.1)$$

with $f(b,t) = 2\arctan\frac{d+b}{t} + 2\arctan\frac{d-b}{t} - \arctan\frac{2d+b}{t} - \arctan\frac{2d-b}{t}$. I majorise $f \le U$ on $|b| \le \frac{1}{2}$, $t \ne 0$, by $U = \sum_\iota c_\iota(F_\iota(d+b,t) + F_\iota(d-b,t))$, with $F_{\mathrm{R},j} = \mathrm{Re}(x+it)^{-j}$ $(1 \le j \le m)$ and $F_{\mathrm{I},j} = -\mathrm{Im}(x+it)^{-j}$ $(2 \le j \le m)$. The sums of U over the zeros are evaluated through ξ'/ξ and its derivatives at $\frac{1}{2}+d+iT$.

1. Optimal kernels. For fixed d, the condition $f \le U$ is linear in the coefficients. The total bound at a target height L^* is a convex function of them, since the Euler-product term below is a sum of Euclidean norms. I minimise $A_1 L^* + A_3$ by second-order cone programming, with cutting planes for $f \le U$ and a margin $\eta/(1+t^2)$, for $d \in \{0.65, \dots, 1.2\}$ and $m \le 5$. The seven lines of (1.1) are the optima for $L^* \in \{5, 20, 60, 150, 400, 1000, 2500\}$. For instance, the line $0.09349L + 3.237$ has $d = 0.85$ and $m = 5$.

2. The Euler product, combined. For each prime power n, the contribution of ζ to the bound is $\alpha_n \cos(T\log n) + \beta_n \sin(T\log n)$. Here α_n collects the real-part terms and β_n the imaginary-part terms and the two values of $\arg\zeta$. I bound it by $\sqrt{\alpha_n^2+\beta_n^2}$, which gives

$$\big|N(T) - \mathrm{M}(T)\big| \le A_1 \log T + C_\zeta + \sup_{T \ge 1} F(T), \qquad C_\zeta = \sum_n \Lambda(n)\, n^{-\sigma_1}\sqrt{a(\log n)^2 + b(\log n)^2}, \quad (2.2)$$

with $\sigma_1 = \frac{1}{2} + d$ and explicit functions a, b. C_ζ is summed exactly in Arb over all prime powers up to 10^8. The tail is bounded by partial summation with $\psi(x) < 1.03883x$ and a non-increasing majorant of the summand. For your kernel (2.43) this gives $C_\zeta < 4.40867$, which corresponds to $C_E/2 \le 4.584$ in your notation.

3. The elementary part, exactly. $F(T)$ collects the contributions of Γ, of $s(s-1)$ and of the main term, minus $A_1 \log T$. I evaluate it as a function of T and take one supremum. On $[1, 10^4]$ this uses a centred form on 9215 intervals. For $T \ge 10^4$ I use Stirling's formula with Olver's remainder [5, 5.11(ii)] and Cauchy estimates for the polygamma functions. For (2.43), $\sup F \le 0.72834$, attained at $T = 1$; for comparison, $(G_u + C_3)/2 \approx 1.16$. In the version for $S(T)$, the constant $\frac{7}{8}$ cancels and $\sup F$ is negative, about -0.16.

4. The kernel inequality. $f \le U$ is certified on the whole half-strip $0 \le b \le \frac{1}{2}$, $t \ne 0$, by centred-form bisection. In a neighbourhood of $t = 0$ I use the form of f that is continuous up to $t = 0^{\pm}$. For $|t| \ge 100$ I use the variable $u = 1/t$, together with the expansion $t^2 f = 4d^3 u + O(u^3)$ with an explicit remainder.

5. The hybrid. By the maximum principle, the constraints that fix the optimum of the kernel are those at $b = \pm\frac{1}{2}$, that is, at hypothetical zeros near $\beta = 0$ or $\beta = 1$. If the constraint is imposed only at $b = 0$, as would suffice if all zeros were on the critical line, the optimal coefficient decreases towards 0 with m: 0.050, 0.020, 0.0126 for $m = 4, 10, 20$. In the hybrid, the kernel may fail near $b = \pm\frac{1}{2}$. The excess is paid by weights $\kappa_k \log^+(R_k/|\rho^* - c_k - i(T+v_k)|)$, where ρ^* is the zero of real part $\ge \frac{1}{2}$ among ρ and $1-\bar\rho$, on discs with $c_k - R_k \ge \frac{1}{2}$. Their sums over the zeros are bounded by Jensen's formula and Bourgain's exponent pair [4], through $2(J_k + \varepsilon)\log T$ with $J_k = \frac{1}{2\pi}\int_0^{2\pi} \mathfrak{e}(c_k + R_k\cos\theta)\, d\theta$. This gives

$$\mathfrak{a} \le \frac{c_{\mathrm{R},1}}{2\pi} + \frac{1}{\pi}\sum_k \kappa_k J_k,$$

and (1.4) for a kernel with $m = 10$, $d = \frac{9}{10}$ and 39 discs.

3. Comparison and attribution

	previous	this letter
$\lvert N - \mathrm{M}\rvert$, $T \geq 1$	$\min(0.0946L + 5.746,\ 0.097L + 4.954)$ [3]	(1.1), below by ≥ 1.996
$\lvert S(T)\rvert$, $T \geq e$	$0.0946L + 6.691$ (from [3])	$0.09349L + 2.350$; (1.2)
final slope, explicit	0.0946	0.08844
$\mathfrak{a}$	0.0946	0.07755 (asymptotic only)

The identity (2.1) and the idea of the kernel majorant are those of [1, 3]. Your §3 in [3] anticipates that more parameters and convex optimisation should lower the leading constant. Items 1–3 of Section 2 carry this out and sharpen the bookkeeping. Item 5, the Jensen discs controlled by subconvexity, is, as far as I know, new.

The programs, the coefficients (as exact binary numbers) and the outputs of all certificates are public at `https://github.com/vcasana1/PaperMath/tree/explicit-S-turing/numerics/explicit_S_turing`.

I would be happy to share more details, and I would greatly appreciate your opinion, in particular on whether you plan to pursue the optimisation in §3 of [3], so that we can avoid duplicating work.

With kind regards,

Víctor Casaña-Giner

References

[1] V. S. G. Amberger, Estimating the number of zeros of Dedekind zeta-functions, preprint, arXiv:2510.27444 (v1 2025, v4), doi:10.48550/arXiv.2510.27444. (Cited on pp. 1, 2, and 3.)

[2] C. Bellotti and P.-J. Wong, Improved estimates for the argument and zero-counting function of the Riemann zeta-function, *Math. Comp.* (2025, published online 14 November 2025), doi:10.1090/mcom/4133. (Cited on p. 1.)

[3] C. Bellotti and P.-J. Wong, Counting zeros of Artin L-functions, preprint, arXiv:2606.31441v1, 2026, doi:10.48550/arXiv.2606.31441. (Cited on pp. 1, 2, and 3.)

[4] J. Bourgain, Decoupling, exponential sums and the Riemann zeta function, *J. Amer. Math. Soc.* **30** (2017) 205–224, doi:10.1090/jams/860. (Cited on p. 2.)

[5] NIST Digital Library of Mathematical Functions, §5.11, `https://dlmf.nist.gov/5.11`. (Cited on p. 2.)

[6] F. Johansson, Arb: efficient arbitrary-precision midpoint-radius interval arithmetic, *IEEE Trans. Comput.* **66** (2017) 1281–1292, doi:10.1109/TC.2017.2690633. (Cited on p. 1.)

[7] J. B. Rosser and L. Schoenfeld, Approximate formulas for some functions of prime numbers, *Illinois J. Math.* **6** (1962) 64–94, doi:10.1215/ijm/1255631807. (Cited on p. 1.)

[8] A. M. Turing, Some calculations of the Riemann zeta-function, *Proc. London Math. Soc. (3)* **3** (1953) 99–117, doi:10.1112/plms/s3-3.1.99. (Cited on p. 1.)